The Hellings–Downs curve and its variance

Derivation notes on Romano & Allen (2024) and Allen (2023)

These notes work through the main calculations in two papers:

- J. D. Romano and B. Allen, *Answers to frequently asked questions about the pulsar timing array Hellings and Downs curve*, Class. Quantum Grav. **41**, 175008 (2024), arXiv:2308.05847.
- B. Allen, *Variance of the Hellings-Downs correlation*, Phys. Rev. D **107**, 043018 (2023), arXiv:2205.05637.

1 Plane waves and polarization tensors

Far from its source, a gravitational wave (GW) near the Earth and the pulsars is a plane wave travelling in direction $\hat{\Omega}$. The source is in direction $-\hat{\Omega}$. In transverse-traceless coordinates

$$h_{ij}(t,\vec{x}) = \sum_{A=+,\times} h^A(t-\hat{\Omega}\cdot\vec{x})\, e^A_{ij}(\hat{\Omega}) = \Re\big[h(t-\hat{\Omega}\cdot\vec{x})\, e^*_{ij}(\hat{\Omega})\big], \tag{1}$$

with the complex combinations $h = h^+ + ih^\times$ and $e_{ij} = e^+_{ij} + ie^\times_{ij}$. The polarization tensors are built from two unit vectors $\hat{m}, \hat{n}$ orthogonal to $\hat{\Omega}$ and to each other:

$$e^+_{ij} = \hat{m}_i\hat{m}_j - \hat{n}_i\hat{n}_j, \qquad e^\times_{ij} = \hat{m}_i\hat{n}_j + \hat{n}_i\hat{m}_j. \tag{2}$$

The second form of (1) follows from $\Re[(h^+ + ih^\times)(e^+ - ie^\times)] = h^+e^+ + h^\times e^\times$.

Following Romano & Allen, take (θ,ϕ) to be the sky position of the source, so

$$\hat{\Omega} = -\hat{r}, \qquad \hat{m} = -\hat{\phi}, \qquad \hat{n} = -\hat{\theta}, \tag{3}$$

where $\hat{r}, \hat{\theta}, \hat{\phi}$ are the usual spherical unit vectors.

Polarization angle. Rotating $\hat{m}, \hat{n}$ by an angle ψ in the plane orthogonal to $\hat{\Omega}$, $\hat{m}' = \cos\psi\,\hat{m} + \sin\psi\,\hat{n}$ and $\hat{n}' = -\sin\psi\,\hat{m} + \cos\psi\,\hat{n}$, gives

$$\hat{m}'\hat{m}' - \hat{n}'\hat{n}' = \cos 2\psi\,(\hat{m}\hat{m} - \hat{n}\hat{n}) + \sin 2\psi\,(\hat{m}\hat{n} + \hat{n}\hat{m}), \tag{4}$$

$$\hat{m}'\hat{n}' + \hat{n}'\hat{m}' = -\sin 2\psi\,(\hat{m}\hat{m} - \hat{n}\hat{n}) + \cos 2\psi\,(\hat{m}\hat{n} + \hat{n}\hat{m}), \tag{5}$$

using $\cos^2\psi - \sin^2\psi = \cos 2\psi$ and $2\sin\psi\cos\psi = \sin 2\psi$. In complex form this is $e'_{ij} = e^{-2i\psi}e_{ij}$. The metric is unchanged if $h' = e^{-2i\psi}h$ at the same time. Quantities that are invariant under ψ are the physical ones; this "spin-two" phase will reappear in the two-point function of Sec. 6.

Unpolarized sources. A source is called unpolarized if, over the observation time T,

$$\overline{(h^+)^2} = \overline{(h^\times)^2}, \qquad \overline{h^+h^\times} = 0. \tag{6}$$

A circularly polarized monochromatic wave $h = Ae^{i(2\pi ft+\phi_0)}$ satisfies this when fT is an integer, because then $\overline{\cos^2} = \overline{\sin^2} = 1/2$ and $\overline{\sin\cos} = 0$. So in this sense "unpolarized" means "no preferred polarization angle", and a circularly polarized source qualifies.

2 Redshift of one pulsar

2.1 From the metric to the redshift

A pulse leaves a pulsar at distance L in direction $\hat{p}$ and travels to the Earth. Parametrize the path by $s \in [0, L]$:

$$t(s) = t - (L - s), \qquad \vec{x}(s) = (L - s)\hat{p}. \tag{7}$$

The extra light travel time caused by the wave is the integral of the metric perturbation projected on the path,

$$\Delta T(t) = \frac{1}{2}\hat{p}^i\hat{p}^j \int_0^L ds\; h_{ij}\big(t(s), \vec{x}(s)\big). \tag{8}$$

For a plane wave the argument of h^A is

$$u(s) = t(s) - \hat{\Omega}\cdot\vec{x}(s) = t - (L - s)(1 + \hat{\Omega}\cdot\hat{p}), \qquad \frac{du}{ds} = 1 + \hat{\Omega}\cdot\hat{p}. \tag{9}$$

The redshift is $Z = d\Delta T/dt$. Because $\partial_t h^A(u) = dh^A/du$, the s-integral is an exact derivative:

$$\int_0^L ds\,\frac{dh^A}{du} = \frac{1}{1 + \hat{\Omega}\cdot\hat{p}}\int_{u(0)}^{u(L)} dh^A = \frac{h^A(t) - h^A\big(t - L(1 + \hat{\Omega}\cdot\hat{p})\big)}{1 + \hat{\Omega}\cdot\hat{p}}. \tag{10}$$

Therefore

$$\boxed{\begin{gathered} Z(t) = \sum_A F^A(\hat{\Omega})\,\Delta h^A(t), \qquad F^A(\hat{\Omega}) = \frac{1}{2}\frac{\hat{p}^i\hat{p}^j e^A_{ij}(\hat{\Omega})}{1 + \hat{\Omega}\cdot\hat{p}}, \\ \Delta h^A(t) = h^A(t) - h^A\big(t - L(1 + \hat{\Omega}\cdot\hat{p})\big). \end{gathered}} \tag{11}$$

The first term of Δh^A is the *Earth term*, the second the *pulsar term*. The pulsar term is the wave as it passed the pulsar, delayed by $\tau = L(1 + \hat{\Omega}\cdot\hat{p})$.

2.2 The size of the antenna pattern

Write $u = \hat{\Omega}\cdot\hat{p}$. Because $\hat{m}, \hat{n}, \hat{\Omega}$ form an orthonormal basis, $(\hat{m}\cdot\hat{p})^2 + (\hat{n}\cdot\hat{p})^2 = 1 - u^2$. Then

$$(F^+)^2 + (F^\times)^2 = \frac{\frac{1}{4}\big[(\hat{m}\cdot\hat{p})^2 - (\hat{n}\cdot\hat{p})^2\big]^2 + (\hat{m}\cdot\hat{p})^2(\hat{n}\cdot\hat{p})^2}{(1+u)^2} = \frac{\frac{1}{4}\big[(\hat{m}\cdot\hat{p})^2 + (\hat{n}\cdot\hat{p})^2\big]^2}{(1+u)^2} = \frac{(1-u^2)^2}{4(1+u)^2}, \tag{12}$$

so

$$|F(\hat{\Omega})| \equiv \sqrt{(F^+)^2 + (F^\times)^2} = \frac{1}{2}\,(1 - \hat{\Omega}\cdot\hat{p}). \tag{13}$$

With $-\hat{\Omega}\cdot\hat{p} = \cos\theta$, where θ is the angle between the pulsar and the source, $|F| = \frac{1}{2}(1 + \cos\theta)$. It is largest for a source behind the pulsar and vanishes for a source in the opposite direction. The denominator $1 + \hat{\Omega}\cdot\hat{p}$ looks dangerous at $\hat{\Omega} = -\hat{p}$ but the numerator vanishes faster.

2.3 A monochromatic circular wave

For $h(t) = Ae^{i(2\pi ft + \phi_0)}$, write $F = F^+ + iF^\times$. Using $Z = \Re[\Delta h\, F^*]$,

$$Z(t) = \Re\Big(Ae^{i(2\pi ft + \phi_0)}\,\mathcal{T}\,F^*\Big), \qquad \mathcal{T}(\alpha, u) = 1 - e^{-2\pi i\alpha(1+u)}, \qquad \alpha = fL = L/\lambda. \tag{14}$$

$\mathcal{T}$ is the redshift transfer function: Earth term minus pulsar term. The redshift oscillates as $r\cos(2\pi ft + \phi)$ with amplitude $r = A|\mathcal{T}||F|$, and $|\mathcal{T}|^2 = 2 - 2\cos[2\pi\alpha(1+u)]$. It vanishes when the pulse crosses a whole number of GW cycles, $1 + u = n/\alpha$. For a typical pulsar $\alpha \sim 10^2$–10^3, so $|\mathcal{T}|$ oscillates rapidly over the sky and $|F|$ is half its envelope.

Short-arm limit. If instead $\alpha \ll 1$, expanding $\mathcal{T} \simeq 2\pi i\alpha(1+u)$ cancels the denominator of F^A:

$$R^A \equiv \mathcal{T}F^A \simeq 2\pi i f L\, G^A, \qquad G^A = \tfrac{1}{2}\hat{p}^i\hat{p}^j e^A_{ij}. \tag{15}$$

G^A is symmetric under $\hat{p} \to -\hat{p}$, as for a LIGO arm. F^A is not, and this asymmetry is behind several features of the HD curve below.

2.4 Average over polarization angle

For a single source with complex waveform h, $Z = \Re(C)$ with $C = \Delta h\, F^*$. Rotating the source's polarization axes multiplies C by $e^{2i\psi}$, so $Z = C_R \cos 2\psi - C_I \sin 2\psi$ and

$$\frac{1}{2\pi}\int_0^{2\pi} d\psi\, Z^2 = \frac{1}{2}|C|^2. \tag{16}$$

The rms redshift of a randomly oriented source is therefore $|\Delta h|\,|F|/\sqrt{2}$. This is why the quadrature combination $|F|$ describes unpolarized sources.

2.5 Many plane waves

A general background is a superposition of plane waves,

$$h_{ij}(t,\vec{x}) = \sum_A \int df \int d\hat{\Omega}\; h_A(f,\hat{\Omega})\, e^A_{ij}(\hat{\Omega})\, e^{2\pi i f(t-\hat{\Omega}\cdot\vec{x})}, \tag{17}$$

and (11) applied to each wave gives

$$Z_a(t) = \sum_A \int df \int d\hat{\Omega}\; R^A_a(f,\hat{\Omega})\, h_A(f,\hat{\Omega})\, e^{2\pi i f t}, \qquad R^A_a = \left[1 - e^{-2\pi i f L_a(1+\hat{\Omega}\cdot\hat{p}_a)}\right] F^A_a(\hat{\Omega}). \tag{18}$$

3 The Hellings–Downs curve

3.1 Definition

The correlation of two pulsars a, b is the time average $\rho_{ab} = \overline{Z_a(t)Z_b(t)}$. For one unpolarized source in direction $-\hat{\Omega}$, each Z has an Earth and a pulsar term for each polarization. Averaging over source directions (Hellings and Downs' original procedure) kills every product that contains a pulsar term of a distinct, distant pulsar: those terms carry phases $2\pi f L(1+\hat{\Omega}\cdot\hat{p})$ that vary over many cycles across the sky, and they enter only through the autocorrelation function of the wave at lags τ_a, τ_b or $\tau_a - \tau_b$, which is small when the lags are long compared with the coherence time. For $a = b$ the pulsar-pulsar term has zero lag and survives, doubling the Earth-Earth term. The result is

$$\langle\rho_{ab}\rangle = h^2\,\mu_{\rm u}(\gamma)\,[1+\delta_{ab}], \qquad h^2 = \tfrac{1}{2}\overline{|h|^2}, \qquad \mu_{\rm u}(\gamma) = \frac{1}{4\pi}\int d\hat{\Omega}\sum_A F^A_a(\hat{\Omega})F^A_b(\hat{\Omega}), \tag{19}$$

where $\cos\gamma = \hat{p}_a\cdot\hat{p}_b$. The cross-polarization terms drop because $\overline{h^+h^\times} = 0$. By rotational symmetry $\mu_{\rm u}$ can depend only on γ.

3.2 The integrand

Summing over polarizations uses the completeness relation for the two tensors,

$$\sum_A e^A_{ij}e^A_{kl} = P_{ik}P_{jl} + P_{il}P_{jk} - P_{ij}P_{kl}, \qquad P_{ij} = \delta_{ij} - \hat{\Omega}_i\hat{\Omega}_j, \tag{20}$$

which can be checked by contracting both sides with $\hat{m}, \hat{n}$ in all combinations. Contracting with $\hat{p}_a^i \hat{p}_a^j \hat{p}_b^k \hat{p}_b^l$, and writing $u_a = \hat{\Omega}\cdot\hat{p}_a$, $u_b = \hat{\Omega}\cdot\hat{p}_b$, $c = \cos\gamma$,

$$\hat{p}_a \cdot P \cdot \hat{p}_b = c - u_a u_b, \qquad \hat{p}_a \cdot P \cdot \hat{p}_a = 1 - u_a^2, \tag{21}$$

so

$$Q(\hat{\Omega}) \equiv \sum_A F_a^A F_b^A = \frac{2(c - u_a u_b)^2 - (1-u_a^2)(1-u_b^2)}{4(1+u_a)(1+u_b)}. \tag{22}$$

3.3 Doing the integral

Put pulsar a on the z-axis and pulsar b in the xz-plane: $\hat{p}_a = \hat{z}$, $\hat{p}_b = (\sin\gamma, 0, \cos\gamma)$. Use $t \equiv u_a$ and the azimuth φ of $\hat{\Omega}$ about $\hat{z}$, so that

$$u_b = tc + \sqrt{1-t^2}\,\sin\gamma\,\cos\varphi, \qquad \frac{1}{4\pi}\int d\hat{\Omega} = \frac{1}{2}\int_{-1}^{1} dt\,\frac{1}{2\pi}\int_0^{2\pi} d\varphi. \tag{23}$$

The φ average. The numerator of (22) is quadratic in $\cos\varphi$ and the only φ-dependent factor in the denominator is $1 + u_b$. Dividing the numerator by $1 + u_b$ (as a polynomial in $\cos\varphi$) leaves the remainder $2(c+t)^2$, which does not depend on φ. The polynomial part averages with $\langle\cos\varphi\rangle = 0$, $\langle\cos^2\varphi\rangle = \frac{1}{2}$, and the remainder needs

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{d\varphi}{A + B\cos\varphi} = \frac{1}{\sqrt{A^2 - B^2}}, \qquad A = 1 + tc, \quad B = \sqrt{1-t^2}\,\sin\gamma. \tag{24}$$

Here a small miracle happens: $(1+tc)^2 - (1-t^2)(1-c^2) = (t+c)^2$, so $\sqrt{A^2-B^2} = |t+c|$. Collecting terms,

$$\frac{1}{2\pi}\int_0^{2\pi} d\varphi\, Q = \frac{c\,t^3 - 3c\,t - t^2 - 1}{4(1+t)} + \frac{|t+c|}{2(1+t)}. \tag{25}$$

The absolute value splits the remaining integral at $t = -c$. The apparent pole at $t = -1$ cancels between the two terms.

The t integral. For $t < -c$ the two terms combine into a polynomial, and for $t > -c$ one term $\propto 1/(1+t)$ is left:

$$t < -c: \quad \tfrac{1}{4}(1+t)(c\,t - 2c - 1), \tag{26}$$

$$t > -c: \quad \tfrac{1}{4}\big(c\,t^2 - c\,t - 2c - t + 3\big) + \frac{c-1}{1+t}. \tag{27}$$

The polynomial pieces integrate to elementary expressions, and $\int_{-c}^{1} dt/(1+t) = \ln[2/(1-c)]$. Adding everything and multiplying by $\frac{1}{2}$,

$$\mu_{\rm u}(\gamma) = \frac{1}{4} + \frac{\cos\gamma}{12} + \frac{1-\cos\gamma}{2}\ln\frac{1-\cos\gamma}{2}. \tag{28}$$

With $x \equiv (1-\cos\gamma)/2$ this is the familiar form

$$\boxed{\mu_{\rm u}(\gamma) = \frac{1}{3} - \frac{x}{6} + x\ln x, \qquad x = \frac{1-\cos\gamma}{2}.} \tag{29}$$

This is the result quoted in Romano & Allen, Sec. 3.1, and Allen (2023) eq. (2.4).

3.4 Properties

Normalization, 1/3 or 1/2. At $\gamma = 0$, $x = 0$ and $\mu_{\rm u} = 1/3$. Two pulsars in the same direction at different distances have $\langle\rho\rangle = h^2/3$; the same pulsar with itself has twice that, because of the δ_{ab} term. If one wants $\mu_{\rm u}$ to read as a correlation coefficient equal to 1 for a pulsar with itself, one multiplies by 3/2 and gets 1/2 at $\gamma = 0$. The choice is a convention and changes nothing observable.

Why $\mu_{\rm u}(180°) = \frac{1}{2}\mu_{\rm u}(0°)$. Take $\hat{p}_a = \hat{z}$ and write $u = \cos\theta$ for the source direction. With both pulsars on $\hat{z}$, $Q = \frac{1}{4}(1+u)^2$ by (13). With $\hat{p}_b = -\hat{z}$, eq. (22) with $c = -1$, $u_b = -u_a$ gives $Q = \frac{1}{4}(1-u^2)$. The sky average of a function of u is $\frac{1}{2}\int_{-1}^{1} du$, so

$$\mu_{\rm u}(0) = \frac{1}{8}\int_{-1}^{1}(1+u)^2 du = \frac{1}{3}, \qquad \mu_{\rm u}(\pi) = \frac{1}{8}\int_{-1}^{1}(1-u^2)du = \frac{1}{6}. \tag{30}$$

A quadrupole deformation treats $\hat{p}$ and $-\hat{p}$ alike; the difference comes entirely from the denominators $1 + \hat{\Omega}\cdot\hat{p}$, i.e. from the pulse travelling *with* or *against* the wave.

Legendre expansion. Expand $\mu_{\rm u} = \sum_l (2l+1) c_l P_l(\cos\gamma)$. By orthogonality, $\int_{-1}^{1} P_l P_{l'} = 2\delta_{ll'}/(2l+1)$, the coefficients are $c_l = \frac{1}{2}\int_{-1}^{1} \mu_{\rm u}\, P_l\, d(\cos\gamma)$. Using Rodrigues' formula and integrating by parts (or simply evaluating the integrals),

$$c_0 = c_1 = 0, \qquad c_l = \frac{1}{(l+2)(l+1)l(l-1)} \quad (l \geq 2), \tag{31}$$

e.g. $c_2 = 1/24$, $c_3 = 1/120$, $c_4 = 1/360$. There is no monopole and no dipole, the quadrupole dominates, and the c_l fall like l^{-4}, so a few terms give a very good approximation. As a check, $\sum_{l\geq 2}(2l+1)c_l = 1/3 = \mu_{\rm u}(0)$.

The minimum is not at 90°. A pure quadrupole $P_2(\cos\gamma)$ would have its minimum at 90°. The odd-l terms shift it. From (29), $d\mu_{\rm u}/dx = -\frac{1}{6} + \ln x + 1 = 0$ gives $x = e^{-5/6}$, i.e.

$$\cos\gamma_{\rm min} = 1 - 2e^{-5/6}, \qquad \gamma_{\rm min} \simeq 82.5°, \qquad \mu_{\rm u}(\gamma_{\rm min}) \simeq -0.101. \tag{32}$$

Short arms. Repeating (19) with G^A instead of F^A gives the sky average $\frac{2}{15}P_2(\cos\gamma)$: a pure quadrupole, symmetric about 90°. Pulsars are "long-arm" detectors, $L \gg \lambda$, and that is the whole difference.

No frequency dependence. The frequency enters R_a^A only through the pulsar-term phase $2\pi f L_a(1 + \hat{\Omega}\cdot\hat{p}_a)$, and those terms drop out of the expected correlation. For two ground-based interferometers the response carries $e^{-2\pi i f \hat{\Omega}\cdot\vec{r}_a}$ with a separation $|\vec{r}_a - \vec{r}_b|$ comparable to the wavelength, so their overlap function depends on frequency. For a PTA the Earth terms of both pulsars are measured at the same place, the Earth, so this factor is 1.

4 Pulsar averaging

In practice one does not average over many universes. One averages the measured correlation over many pulsar pairs with the same separation γ:

$$\Gamma(\gamma) \equiv \langle\rho_{ab}\rangle_{ab\in\gamma}. \tag{33}$$

For a single source this gives the same curve. The integrand $F_a(\hat{\Omega})F_b^*(\hat{\Omega})$, with $F = F^+ + iF^\times$, is unchanged when the pulsar pair and the source (with its polarization frame) are rotated

together: the spin-two phases $e^{\mp 2i\psi}$ of F_a and F_b^* cancel. Averaging over all rotations of the pulsar pair at fixed source is therefore the same as averaging over all rotations of the source at fixed pulsars, which is a uniform average over $\hat{\Omega}$ and over ψ. Since $\Re(F_a F_b^*) = \sum_A F_a^A F_b^A$, this reproduces $\mu_{\rm u}(\gamma)$. Pulsar averaging also removes the pulsar terms, provided pulsars in the same direction are many wavelengths apart in distance.

5 One point source: the pulsar variance

Fix one unpolarized source. Different pulsar pairs with the same γ see different correlations; their spread is the *pulsar variance*. Neglecting pulsar terms, the correlation is $\rho = Q(\hat{\Omega})$ of (22), and by the argument of the previous section the spread over pulsar pairs equals the spread over source directions:

$$\sigma_{\rm u}^2(\gamma) = \langle Q^2\rangle - \langle Q\rangle^2, \qquad \langle\cdot\rangle = \frac{1}{4\pi}\int d\hat{\Omega}. \tag{34}$$

The second moment is the same kind of integral as before, with Q^2 in place of Q; it is longer but contains nothing new. Allen (2023) eq. (2.6) gives

$$\sigma_{\rm u}^2(\gamma) = \frac{97}{80} + \frac{\cos\gamma}{24} - \frac{839}{720}\cos^2\gamma + \frac{1-\cos\gamma}{12}\,\ln x\left[18 - 10\cos\gamma - 3(1-\cos\gamma)\ln x\right], \tag{35}$$

with $x = (1-\cos\gamma)/2$. The two ends are quick to check by hand:

$$\gamma = 0: \quad \langle Q^2\rangle = \frac{1}{2}\int_{-1}^{1}\frac{(1+u)^4}{16}du = \frac{1}{5}, \qquad \sigma_{\rm u}^2 = \frac{1}{5} - \frac{1}{9} = \frac{4}{45}, \tag{36}$$

$$\gamma = \pi: \quad \langle Q^2\rangle = \frac{1}{2}\int_{-1}^{1}\frac{(1-u^2)^2}{16}du = \frac{1}{30}, \qquad \sigma_{\rm u}^2 = \frac{1}{30} - \frac{1}{36} = \frac{1}{180}, \tag{37}$$

and the closed form gives the same values. The pulsar variance is large: at small separations $\sigma_{\rm u} \approx 0.30$ against a mean of $1/3$. A single pulsar pair says little; many pairs are needed.

6 Many sources: interference and the cosmic variance

6.1 The model

Take N point sources in directions $-\hat{\Omega}_j$ with complex waveforms $h_j(t)$. The redshift is the sum of (11) over sources. Keeping only the Earth terms for now,

$$Z_a(t) = \Re\sum_j h_j(t) F_a^*(\hat{\Omega}_j) = \frac{1}{2}\sum_j\left[h_j F_a^*(\hat{\Omega}_j) + h_j^* F_a(\hat{\Omega}_j)\right], \tag{38}$$

and the correlation is a double sum,

$$\rho_{ab} = \frac{1}{4}\sum_{j,k}\Big[\overline{h_j h_k^*}F_a^*(\hat{\Omega}_j)F_b(\hat{\Omega}_k) + \overline{h_j^* h_k}F_a(\hat{\Omega}_j)F_b^*(\hat{\Omega}_k) + \overline{h_j h_k}F_a^*(\hat{\Omega}_j)F_b^*(\hat{\Omega}_k) + \overline{h_j^* h_k^*}F_a(\hat{\Omega}_j)F_b(\hat{\Omega}_k)\Big]. \tag{39}$$

6.2 Sources that do not interfere

If the sources radiate at different (resolvable) frequencies, all time averages with $j \neq k$ vanish. Pulsar-averaging the diagonal terms, $\langle F_a^*(\hat{\Omega}_j)F_b(\hat{\Omega}_j)\rangle_{ab\in\gamma} = \mu_{\rm u}(\gamma)$ and $\langle F_a F_b\rangle_{ab\in\gamma} = 0$, gives

$$\Gamma(\gamma) = \frac{1}{2}\Big(\sum_j \overline{|h_j|^2}\Big)\mu_{\rm u}(\gamma) \tag{40}$$

in *every* universe. With enough pulsars one would recover the HD curve exactly; there is no cosmic variance.

6.3 Interfering sources: the confusion-noise model

PTA sources complete only a few cycles over the observation time, so sources in the same frequency bin do interfere. Allen's "confusion-noise" model gives all sources the same frequency f (a whole number of cycles in T) and circular polarization,

$$h_j(t) = A_j e^{i(2\pi f t + \phi_j)}, \tag{41}$$

with independent phases ϕ_j uniform in $[0, 2\pi)$ and independent directions uniform on the sphere. Then $\overline{h_j^* h_k} = A_j A_k e^{i(\phi_k - \phi_j)}$ and $\overline{h_j h_k} = 0$. Define

$$H_2 = \sum_j A_j^2, \qquad H_4 = \sum_j A_j^4. \tag{42}$$

The two-point function. The pulsar average of $F_a(\hat\Omega_j) F_b^*(\hat\Omega_k)$ for two *different* source directions is the HD two-point function,

$$\langle F_a(\hat\Omega_j) F_b^*(\hat\Omega_k)\rangle_{ab\in\gamma} = \mu(\gamma, \beta_{jk})\, e^{2i\chi_{jk}}, \qquad \cos\beta_{jk} = \hat\Omega_j \cdot \hat\Omega_k, \tag{43}$$

where μ is real and $\chi_{jk} = -\chi_{kj}$ is a spin-two phase that depends on the two source directions. For $\beta = 0$ it reduces to the HD curve, $\mu(\gamma, 0) = \mu_{\rm u}(\gamma)$, and for antipodal sources it vanishes, $\mu(\gamma, \pi) = 0$. Expanding the antenna pattern in spin-weighted spherical harmonics and averaging over pulsar pairs with the addition theorem gives the series

$$\mu(\gamma, \beta) = \sum_{l\geq 2} (2l+1)\, c_l\, P_l(\cos\gamma)\, d^l_{22}(\beta), \tag{44}$$

with the same c_l as in the HD expansion and d^l_{22} the Wigner small-d function, $d^l_{22}(\beta) = \cos^4(\beta/2)\, P^{(0,4)}_{l-2}(\cos\beta)$. Since $d^l_{22}(0) = 1$ and $d^l_{22}(\pi) = 0$, both limits above follow. The two-point function was introduced by Allen (2023, Appendix G) and generalized to arbitrary source directions in later work cited by Romano & Allen (Q7).

The pulsar-averaged correlation. Inserting into (39) and splitting diagonal from off-diagonal terms,

$$\Gamma(\gamma) = \frac{1}{2} H_2\, \mu_{\rm u}(\gamma) + \frac{1}{2} \sum_{j\neq k} A_j A_k \cos\Delta_{jk}\; \mu(\gamma, \beta_{jk}), \qquad \Delta_{jk} = \phi_j - \phi_k - 2\chi_{jk}. \tag{45}$$

The second sum is made of cross-sections of $\mu(\gamma, \beta)$ at $\beta \neq 0$. These do not have the shape of $\mu_{\rm u}(\gamma)$, so in any one universe $\Gamma(\gamma)$ is not proportional to the HD curve, however many pulsars are used.

Mean. Averaging over the random phases, $\langle \cos\Delta_{jk}\rangle = 0$ for $j \neq k$, so

$$\langle \Gamma(\gamma)\rangle = \frac{1}{2} H_2\, \mu_{\rm u}(\gamma). \tag{46}$$

Cosmic variance. Squaring the second sum of (45) needs $\langle \cos\Delta_{jk} \cos\Delta_{lm}\rangle$ for $j \neq k$, $l \neq m$. Writing the cosines as exponentials, only pairings with $\{l, m\} = \{j, k\}$ survive the phase average, and $\Delta_{kj} = -\Delta_{jk}$ because χ is antisymmetric. So

$$\langle \cos\Delta_{jk} \cos\Delta_{lm}\rangle = \tfrac{1}{2}\big(\delta_{jl}\delta_{km} + \delta_{jm}\delta_{kl}\big), \tag{47}$$

and the phases χ drop out. Then

$$\langle\Gamma^2\rangle - \langle\Gamma\rangle^2 = \frac{1}{4}\sum_{j\neq k} A_j^2 A_k^2\, \mu^2(\gamma, \beta_{jk}). \tag{48}$$

Finally average over source directions. For independent uniform directions, $\cos\beta$ is uniform on $[-1, 1]$, so μ^2 is replaced by

$$\tilde{\mu}^2(\gamma) \equiv \frac{1}{2}\int_0^\pi \sin\beta\, d\beta\; \mu^2(\gamma, \beta) = \sum_{l\geq 2} (2l+1)\, c_l^2\, P_l^2(\cos\gamma), \tag{49}$$

where the last step uses (44) and the orthogonality $\int_0^\pi d^l_{22} d^{l'}_{22} \sin\beta\, d\beta = 2\delta_{ll'}/(2l+1)$. With $\sum_{j\neq k} A_j^2 A_k^2 = H_2^2 - H_4$,

$$\boxed{\sigma^2_{\rm cosmic}(\gamma) = \frac{1}{4}\big(H_2^2 - H_4\big)\,\tilde{\mu}^2(\gamma) \;\approx\; \frac{1}{4}H_2^2\,\tilde{\mu}^2(\gamma) \quad (N \gg 1).} \tag{50}$$

This is Allen (2023) eq. (4.8). For many sources $H_4 \ll H_2^2$, and the ratio $\sigma_{\rm cosmic}/\langle\Gamma\rangle = \sqrt{\tilde{\mu}^2}/\mu_{\rm u}$ does not shrink as $N \to \infty$. At $\gamma = 0$, $\tilde{\mu}^2 = \sum(2l+1)c_l^2 = 1/108$, so $\sigma_{\rm cosmic}/\langle\Gamma\rangle = 3/\sqrt{108} \simeq 0.29$.

The random-walk picture explains why: the off-diagonal sum has $N(N-1)$ terms of random sign, so it is of order N, the same order as the diagonal sum.

7 One pulsar pair in a Gaussian ensemble

The total variance of ρ_{ab} for a *single* pair contains both the pulsar and the cosmic variance, and here the pulsar terms matter. With many sources the redshift at frequency f is $Z_a(t) = \Re[c_a e^{2\pi i f t}]$ with

$$c_a = \sum_j A_j e^{i\phi_j}\big(1 - e^{-2\pi i f L_a(1+\hat{\Omega}_j\cdot\hat{p}_a)}\big)F_a^*(\hat{\Omega}_j), \qquad \rho_{ab} = \tfrac{1}{2}\Re\big(c_a c_b^*\big). \tag{51}$$

For large N the c_a are, by the central limit theorem, complex Gaussian with $E[c_a c_b] = 0$ (random phases). Their covariance is

$$K_{ab} \equiv E[c_a c_b^*] = H_2\,\mu_{\rm u}(\gamma) \quad (a \neq b), \qquad K_{aa} = 2H_2\,\mu_{\rm u}(0). \tag{52}$$

For distinct, distant pulsars the pulsar-term pieces average away over source directions. For $a = b$, $|1 - e^{-i\theta}|^2 = 2 - 2\cos\theta$ averages to 2, which is the doubling; and $\langle|F|^2\rangle = \mu_{\rm u}(0) = 1/3$ by (13).

For complex Gaussian variables with $E[c\,c] = 0$, the fourth moments follow from Isserlis' theorem,

$$E\big[|c_a c_b^*|^2\big] = K_{aa}K_{bb} + |K_{ab}|^2, \qquad E\big[(c_a c_b^*)^2\big] = 2K_{ab}^2. \tag{53}$$

With $X = c_a c_b^*$ and K_{ab} real, $E[(\Re X)^2] = \frac{1}{2}E|X|^2 + \frac{1}{2}\Re E[X^2]$, so

$$\mathrm{Var}(\Re X) = \tfrac{1}{2}\big(K_{aa}K_{bb} + K_{ab}^2\big), \qquad \boxed{\mathrm{Var}(\rho_{ab}) = \frac{1}{8}\big(K_{aa}K_{bb} + K_{ab}^2\big) = \frac{1}{8}H_2^2\Big[\mu_{\rm u}^2(\gamma) + 4\mu_{\rm u}^2(0)\Big].} \tag{54}$$

This is Allen (2023) eq. (4.9) in the large-N limit. Without the pulsar terms the $4\mu_{\rm u}^2(0)$ would be $\mu_{\rm u}^2(0)$: the pulsar terms do not change the mean, but they increase the variance of a single pair by a factor of three to four, depending on γ. At every separation the standard deviation is larger than the mean, which is why the measured correlations of individual pairs scatter so widely about the HD curve.